Discrete Maths QUIZ 1 ZSY

Solution 1 (15)

Determine whether $[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$ is a tautology.

ANSWER

WAY 1: using truth table

WAY 2: we use conjunctive normal form

$(p \vee q) \wedge (p \to r) \wedge (q \to r) = (p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee r)$

$p \vee q$				
	0	0	0	0,1
	0	0	1	

$\neg p \vee r$				
	1	0	0	4,6
	1	1	0	

$\neg q \vee r$				
	0	1	0	2,6
	1	1	0	

r				
	0	0	0	0,2,4,6
	0	1	0	
	1	0	0	
	1	1	0	

Left $= \pi(0, 1, 2, 4, 6)$, right $= \pi(0, 2, 4, 6)$.
Therefore Left $\to$ right is a tautology.

Solution 2 (20)

Determine whether each of the following formulae is true, if it is true, prove it; if not, give a counterexample. Where $P(x)$ and $Q(x)$ are predicates, the

domain is positive integer.

a) $\forall x(P(x) \vee Q(x)) = \forall xP(x) \vee \forall xQ(x)$

b) $\exists x(P(x) \vee Q(x)) = \exists xP(x) \vee \exists xQ(x)$

ANSWER

a is false

counterexample: $P(x)$ x is odd $Q(x)$ x is even
then:

- $\forall x(P(x) \vee Q(x))$ is true
- $\forall xP(x)$ is false
- $\forall xQ(x)$ is also false
- $\forall xP(x) \vee \forall xQ(x)$ is also false

b is true

Proof

- if $\exists x(P(x) \vee Q(x)) = T$
- for some c in domain: $P(c) \vee Q(c) = T$,
- then $P(c) = T$ or $Q(c) = T$,
- then $\exists xP(x) = T$ or $\exists xQ(x) = T$
- therefore $\exists xP(x) \vee \exists xQ(x) = T$.

Solution 3 (20)

Suppose that there are three types of people on an island, knights, knaves, and spies. Knights always tell the truth, knaves always lie, and spies sometimes lie and sometimes tell the truth. We know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of the other two is. For each of these situations, if possible, determine who the knave, knight, and spy are:

- A says "I am the knight," B says "A is not the knave," and C says "B is not the knave."
- What types of A, B, C?

ANSWER

A says "I am the knight," B says "A is not the knave," and C says "B is not the knave."

knight	knave	soy	A says	B says	C says
A	B	C	✓	×	✓
A	C	B	✓	✓	×
B	A	C	✓	×	✓
B	C	A	✓	✓	×
C	A	B	✓	×	×
C	B	A	×	✓	×

Solution 4 (20)

Show that the set $\mathbb{Z} \times \mathbb{Z}$ is countable.

ANSWER

Proof 1:
The function $f(m,n) = \frac{(m+n-2)(m+n-1)}{2} + m$ is a one - to - one function from $\mathbb{Z}^+ \times \mathbb{Z}^+$ to $\mathbb{Z}^+$.

Proof 2:
We can list the pairs in the following well - organized way:

$$\begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & \cdots \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & \cdots \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & \cdots \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & \cdots \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Solution 5 (15)

Let $x \in \mathbb{R}$, $f(x) = \lceil x \rceil$, find:

- (1) $f^{-1}(\{1, 2, 3\})$
- (2) $f^{-1}(\{x \mid 1 < x < 2\})$

ANSWER

- $f^{-1}(\{1, 2, 3\}) = (0, 3]$
- $f^{-1}(\{x \mid 1 < x < 2\}) = \varnothing$

Solution 6 (10)

Let $B = \{7k \mid k \in \mathbb{Z}^+\}$, the set of positive integers that are multiples of seven. $A = \mathbb{Z}^+ - B$, constructing a one-to-one correspondence function from A to $\mathbb{Z}^+$.

- $A = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 22, 23, \cdots\}$
- $\mathbb{Z}^+ = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, \cdots\}$

ANSWER

$$F(x) = x - \left\lfloor \frac{x-1}{7} \right\rfloor$$